

Chapter 2

Motivation, Definitions, Terminology

In order to ease our way into the subject of inversive geometry, let us imagine something that is very easy to imagine and something that does not require any specialized mathematical knowledge, a ball on a table.

In other words, as a thought experiment, let us pretend that we put a ball on a leveled table, we wait for that ball to come to a complete stop and now we are looking at a ball that rests on a table.

More technically, let us agree that such a ball represents a sphere, named σ , with the radius R and that such a table represents a plane, named p .

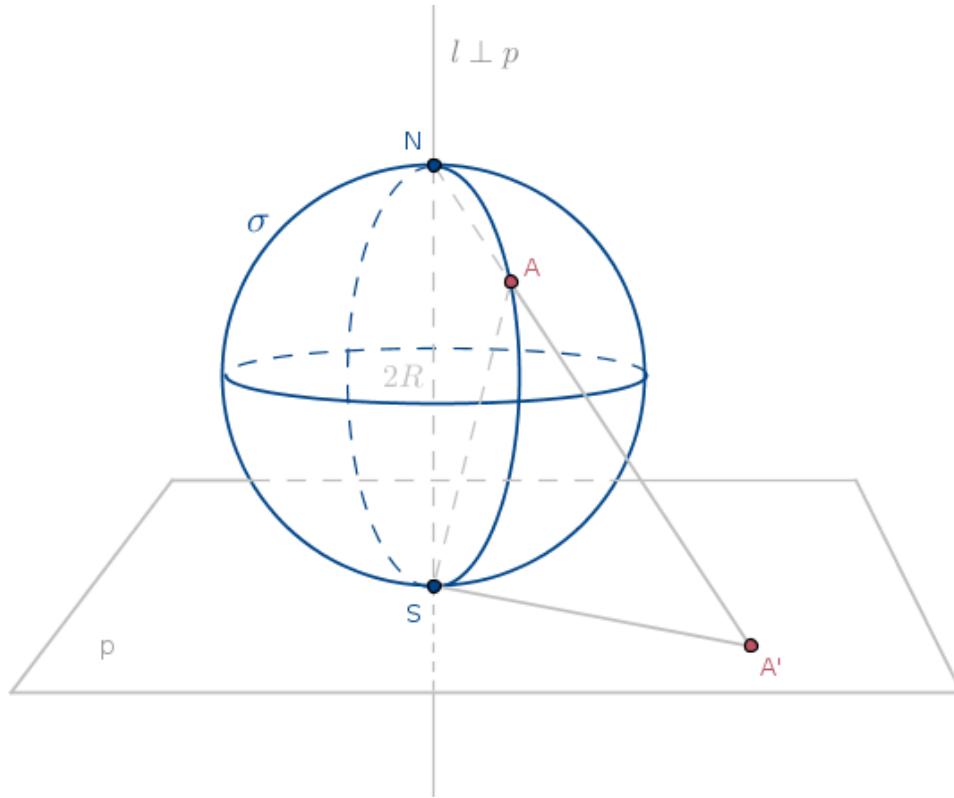


Let us agree further that the point, S , where the sphere, σ , touches the plane, p , will be the sphere's South Pole.

If we draw a straight line, named l , perpendicular to the plane p through the South Pole S then that straight line l will intersect the sphere σ for the first time at the point S and for the second time - at the point N , to which we will refer as the sphere's North Pole.

It should be evident that the line segment SN will be one of the sphere's diameters, meaning that $|SN| = 2R$, and that that diameter will be perpendicular to the plane p .

We now glue or fix the sphere σ to the plane p at the point S , climb on top of that sphere into the point N and standing at the point N we shine a laser light through the sphere onto the plane below (Figure 2.1):



As a result, our laser light will intersect the sphere σ at a point A and the plane p at a point A' .

Technically, our nifty little contraption to which we refer as “a ball on a table” is known in grown-up mathematics as the *Riemann Sphere* and the point A' that we constructed above is known as the *stereographic projection* of the point A .

If we, patiently swinging our laser light up and down, gradually turn around and make one complete revolution about the point N then, in theory, we will probe almost all of the points A that make up our sphere σ and, with the corresponding points A' , we will paint the entire plane p .

By so doing we will create the stereographic projection of the sphere σ onto the plane p .

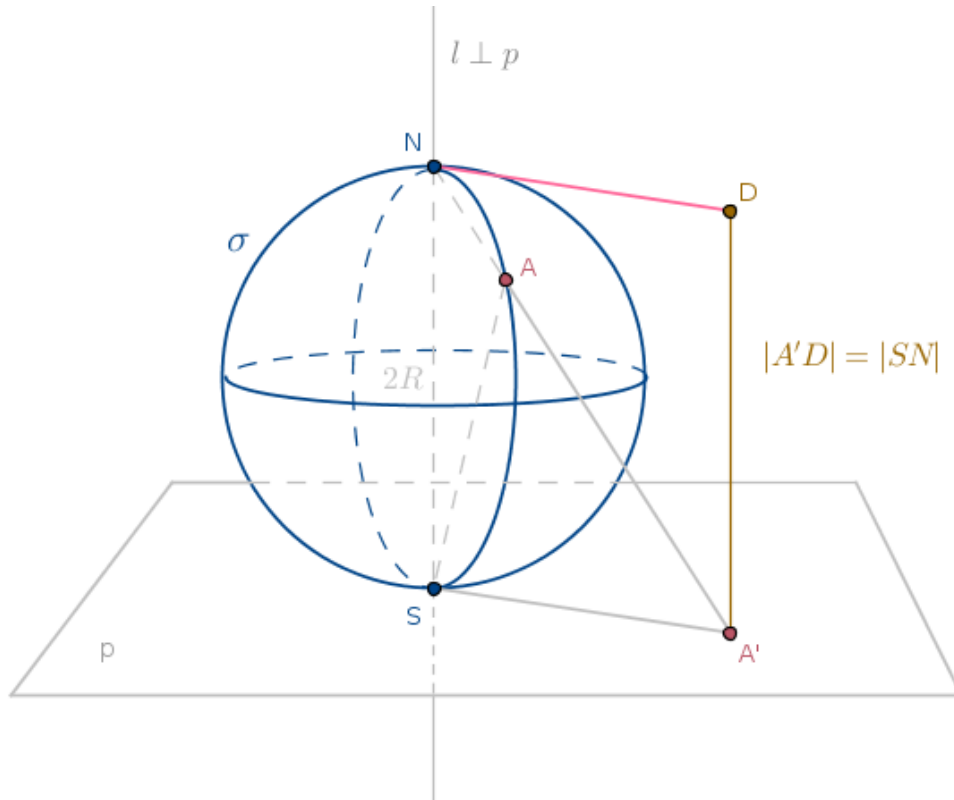
Conversely. We can hop off the sphere onto the plane and roam that plane all over the place shining our laser light at the sphere's North Pole N .

In that case the laser light emitted by the plane p will, before hitting the point N , intersect the sphere σ at a point A that is distinct from N , as in $A \neq N$. It should be fairly evident that as a result of this experiment we will also paint, with the corresponding points A , almost the entire sphere σ .

Why *almost*?

Well, from the way in which we generate our points A on the sphere σ it is clear that no matter how finitely far away from the points S and N we stand, our laser light will never **intersect** that sphere at a point A that will coincide with the sphere's North Pole N .

The only technically sound thing that we can do in this case is jump off of the plane p , hover above that plane at the distance that is equal to the distance, $|SN|$, between the points S and N and then and only then will we be able to smoothly roll the point A directly into the point N in a limiting fashion (Figure 2.2):

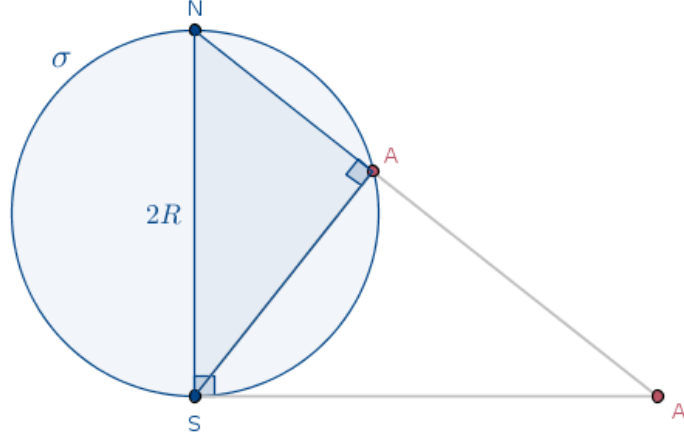


at which point the straight line or the laser light that passes through the points D and N will be tangent to the sphere σ at the point N and that ray of light will be parallel to the plane p .

The trouble is, we are not allowed to jump off of the plane p and conversely.

If we place our laser light into the point N and shine that light through the sphere σ onto the plane p then no matter how close the point A gets to the point N that point N can never be stereographically projected onto the plane p in a genuine way.

In order to see that, let us do two things. First, let us cut our ball-on-a-table contraption with a plane that passes through the points N , S and A' and, second, let us turn the result of that section in such a way that all the interesting material in that section is parallel to the plane of our monitors (Figure 2.3):



Since our cutting plane contains the, not shown above, center of the sphere σ , it follows that the points S , N and A belong to a great circle of that sphere. In that circle the line segment SN is a diameter and the angle NAS subtends that diameter. However, by Euclid's "*Elements*" Book 3 Proposition 31, the angle in a circumference in any circle is right:

$$\angle NAS = \frac{\pi}{2}$$

Hence, the triangle NAS is right.

The triangle NSA' is right by construction.

But these two right triangles share one and the same interior angle at the vertex N . Hence, by Euclid's "*Elements*" Book 1 Proposition 32, the magnitude of the interior angle of the triangle NAS at S and of the triangle NSA' at A' must be equal one another:

$$\angle NSA = \angle NA'S$$

As such, the right triangles NAS and NSA' are similar by **AAA**:

$$\angle NAS = \angle NSA' = \frac{\pi}{2}$$

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$$\angle NSA = \angle NA'S$$

$$\angle SNA = \angle SNA'$$

By Euclid's "*Elements*" Book 6 Proposition 4, in similar triangles the lengths of the sides about equal angles where these sides lie opposite the corresponding equal angles are in the same proportion, the lengths of the sides of our two right triangles about their right angles are in the same proportion:

$$\frac{|SA'|}{|SN|} = \frac{|AS|}{|AN|}$$

from where, for the length of the line segment AN , we have:

$$|AN| = \frac{|AS| \times |SN|}{|SA'|} = \frac{2R \times |AS|}{|SA'|}$$

since $|SN| = 2R$.

If we move the point A over the surface of the sphere σ closer and closer to the sphere's North Pole N then distance, $|AS|$, between the points S and A will be most definitely strictly larger than the radius of that sphere, R :

$$|AS| > R$$

from where, via the last equality, it follows that:

$$|AN| > \frac{2R^2}{|SA'|}$$

which exactly means that as the length of the line segment AN arbitrarily diminishes as the point A gets arbitrarily close to the point N , the length of the line segment SA' arbitrarily grows as the point A' gets arbitrarily far from the point S .

Which, in turn, shows that under the stereographic projection the point N cannot have a genuine image in the plane p .

However, from the topological point of view it would be nice if **every** point of the sphere σ had a stereographic image. To that end, in order to patch things up and extend the set of points on the sphere that have a corresponding image in the plane under the stereographic projection, we introduce a single idealized point known as *the point at infinity* which is designated as P_∞ or as just ∞ .

That idealized point can be vaguely thought of as a point that is "smudged" in the corresponding plane over all possible directions the either lead to or emanate from the point S .

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Slightly more technically, the point at infinity can be represented by any point in the parent plane that is located strictly outside of the circle that has S as its center and any finite real number ρ , no matter how large, as its radius.

In the opposite direction, a tangible image of that single idealized point, the point at infinity, that actually can be drawn is precisely the (single) North Pole of the Riemann sphere, the (single) point N .

The introduction of the point at infinity that plays the role of the stereographic image of the sphere's North Pole in the corresponding plane completes the picture and establishes a topological equivalence between a sphere and the so-called *extended* plane, a plane equipped with the single point at infinity.

The point at infinity plays an important role in Complex Analysis but in this book we will not delve deeper into the properties of that point and we will not be placing too much emphasis on it. We only ask our readers to make a mental note of that notion because we will come back to it soon.

Definitions

Stepping up to the plate, let us reuse the similarity of the right triangles NAS and NSA' but in a slightly different way, by setting up a different proportion. Instead of working with the sides of these triangles that form the respective right angles, let us bring in the hypotenuses of these triangles:

$$\frac{|NA'|}{|NS|} = \frac{|NS|}{|NA|}$$

from where, since $|NS| = 2R$, it follows that:

$$|NA| \times |NA'| = |NS|^2 = 4R^2$$

We, thus, see that under the stereographic projection the product between the distances from the North Pole, N , to a point on the sphere, A , and to its stereographic image, A' , is equal to the square of the parent sphere's diameter.

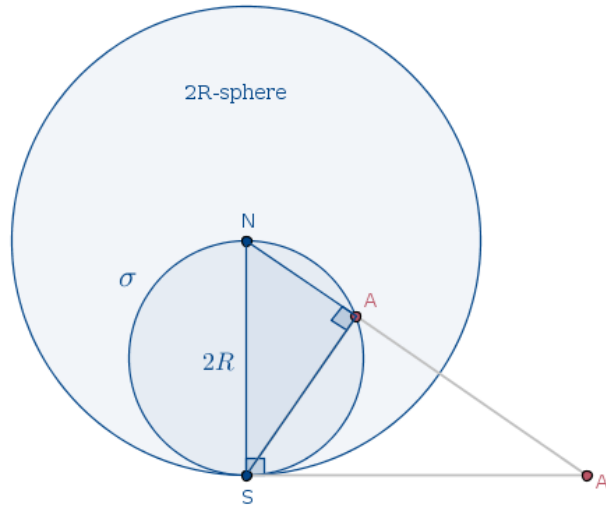
Observe that taking the setup of things shown in Figure 2.3 above as the base, it is possible to paint an infinite straight line that passes through the points S and A' by moving the point A over the periphery of the blue circle that is still designated as σ , so long that we, of course, introduce a similar point at infinity into the works.

That is. If we move the point A over the right semi-circle delineated by the points N and S then we will paint the infinitely long right branch of our straight line SA' and if we wiggle the point A over the left semi-circle delineated by the same points N and S then we will paint the infinitely long left branch of the straight line SA' .

Evidently, in order to make sure that the point N with this mechanism does have a genuine “projective” image on the straight line SA' , we need to introduce a point at infinity that lives, in a strange way, on either side of the point S , at “negative” infinity to, say, left of S and at “positive” infinity to the right of S .

Our goal, however, is to paint not just a single infinite straight line but to paint the entire plane.

To that end, to our current contraption let us add yet another sphere that will have the point N as its center and $2R$ is its radius. Purely for the sake of not cluttering our discussion with too many symbols, let us refer to that sphere as just $2R$ -sphere (Figure 2.4):



From the above figure it should be clear that even though our point A is now located **inside** of the $2R$ -sphere, all the previously constructed and established ratios still hold. By formally replacing the magnitude $2R$ with r , we obtain:

$$|NA| \times |NA'| = r^2$$

Thus, by manipulating our spheres in such a simple way, we observe the following:

- the ray NA connects the center of the $2R$ -sphere with the point A
- the straight line AS is perpendicular to the straight line NA
- the straight line that is perpendicular to the ray NS intersects the ray NA at the point A'

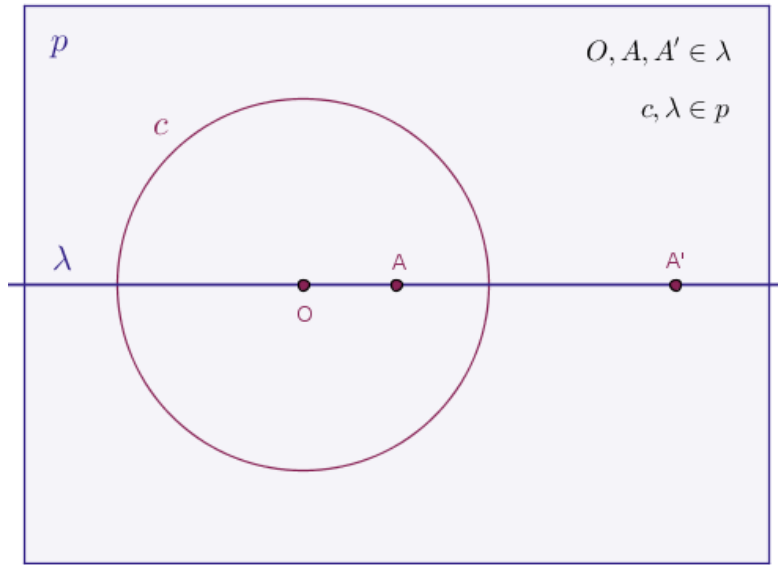
Keep the Figure 2.4 in mind.

What we have just performed is called the inversion of 3-space with respect to a sphere. Such an inversion carries a fair amount of similarity with inversion of 2-space with respect to a circle which we will now take a closer look at.

Inversion With Respect To A Circle In The Real Plane

Definition 1. The inversion of a plane p with respect to a circle c with the center at the point O and the radius r is a transformation of the plane p into itself under which each point A of the plane p is taken into a point A' in such a way that the following three conditions are met.

Condition 1. All three points O , A and A' are collinear and, along with the circle c , are coplanar with the plane p (Figure 2.5):



Condition 2. The points A and A' are distinct from the center of inversion, O , and the point at infinity, P_∞ .

Condition 3. It is the case that $|OA| \times |OA'| = r^2$.

The circle c is called *the circle of inversion*.

The point O is called *the center of inversion*.

The radius r is called *the radius of inversion*.

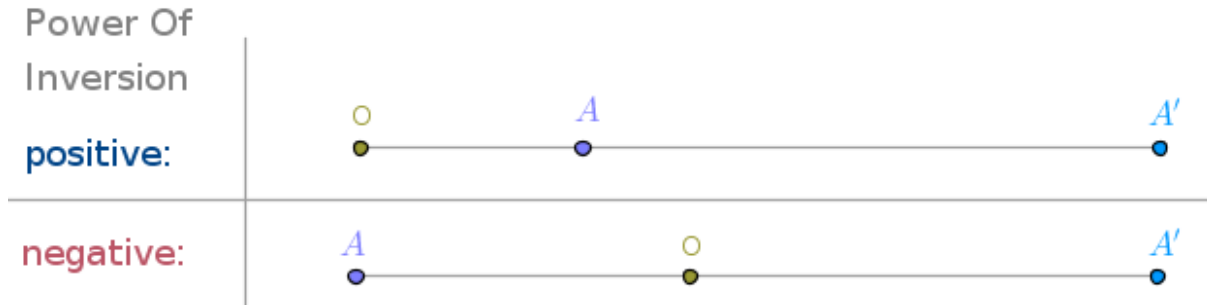
The square of the radius of inversion, r^2 , is called *the power of inversion*.

The power of inversion comes in two distinct flavors or types:

- *positive* and
- *negative*

The type of the power of inversion is the characteristic of the mutual juxtaposition of the points A and A' with respect to the center of inversion.

Definition 2. The power of inversion is *positive*, designated as $+r^2$ or simply as r^2 , if and only if the points A and A' are both located on the same side of the center of inversion (Figure 2.6):



Definition 3. The power of inversion is *negative*, designated with $-r^2$, if and only if the center of inversion is located in between the points A and A' in the sense of the Hilbert's axioms of geometry.

The negative sign in front of the power of inversion is largely symbolic and we will mostly omit it unless required by the context. As such, we will almost exclusively refer to the power of inversion verbally as in "inversion with negative power" or "inversion with positive power".

If the qualifier of the power of inversion is omitted then it means that the inversion in question is applicable to both types of power. We will look at these various powers of inversion in more detail as we develop the subject.

Lastly, under the inversion of a plane with respect to a circle the center of inversion, O , does not really have an image. This follows from Condition 3 of Definition 1 directly if the length of the line segment OA is diminished without bound.

For completeness, though, we shall take it that under the inversion of a plane with respect to a circle with both positive and negative powers the center of inversion is taken into the point at infinity and conversely, the point at infinity is taken into the center of inversion.

Terminology

In the global context of various transformations and mappings in mathematics in general the point A is called *the preimage* and the point A' is called *the image* of the point A under the inversion of the plane with respect to a circle ...

In addition, the point A' may also be referred to as *the inverse image* or as simply *the inverse* of the point A and so on.

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The official, technically correct and faithful but long and, admittedly, cumbersome phrase “*the inversion of the real plane with respect to a circle*” is quickly compressed into “*inversion with respect to a circle*”, which, itself, is compressed further into the equivalent “*inversion in a circle*”.

In order to emphasize the in-spirit similarity of inversion in a circle with the popular geometric transformations of the plane known as *the reflection in a point* and *the reflection in a straight line*, inversion in a circle is sometimes referred to as *the reflection in a circle*, which, as we understand it now, is an abbreviation of the phrase “*the reflection of the real plane in a circle*”.

From the standpoint of the Theory Of Groups, all of the above phrases that deal with the inversion of the plane with respect to a circle have yet another succinct and technically correct incarnation, “*the symmetry of the real plane in a circle*” or just “*symmetry in a circle*”.

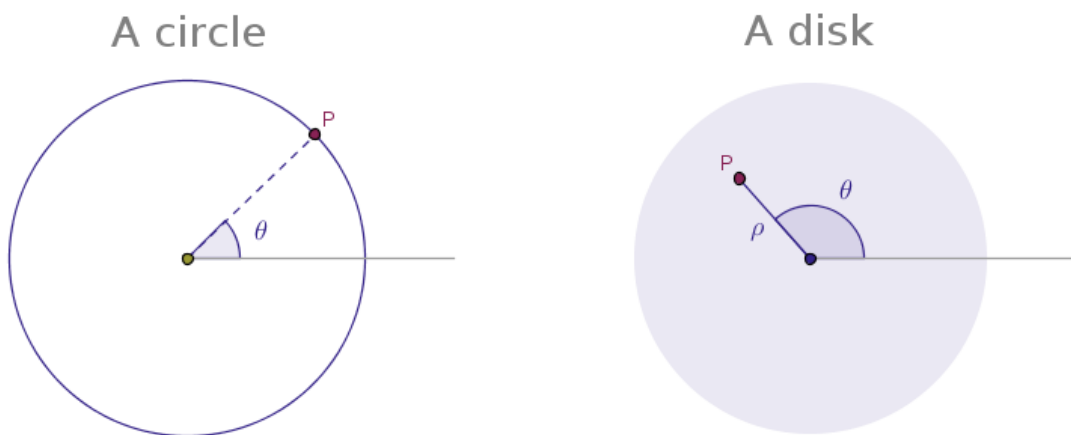
We reserve the right to and will be using all of the above phrases interchangeably.

Common Language

Before we move any further, we would like to clarify an issue that students are frequently confused by, the difference between **a circle** and **a disk** in mathematics.

A circle is a closed planar curve all the points of which have one and the same, fixed, distance from a given, also fixed, point that is located in the same plane as the curve.

Thus, a circle is actually a one-dimensional object because the location of every point, P , that belongs to a circle can be described completely with just one real number, the angular displacement, θ , of that point **on** a circle measured from an arbitrarily chosen axis in an agreed upon direction (Figure 2.7):



A disk is a planar region each point of which is located not further away from a given in the same plane, fixed, point than a given non-negative real number.

Thus, a disk is a two-dimensional object because the location of every point, P , that belongs to a disk can be described completely with two real numbers:

- the angular displacement, θ , of that point **in** a disk measured from an arbitrarily chosen axis in an agreed upon direction and
- the linear displacement, ρ , of that point away from the given fixed point, the center of that disk

The fixed distance in the above definition of a circle is called **the radius** of that circle and the fixed non-negative real number in the above definition of a disk is also called the radius of that disk.

If the radius of a circle and the radius of a disk become zero then both of these objects degenerate into a single geometric point, which is a tactically viable and useful maneuver.

Analytically, an origin-centered circle with the radius r is rendered in a traditional Cartesian coordinate system symbolically as follows:

$$x^2 + y^2 = r^2$$

where the equal sign makes all the difference.

In contrast, the corresponding analytic representation of an origin-centered disk depends on a property of a disk that is not available for a circle, **the closure** or the presence or the absence of **the boundary**.

That is, if a disk of radius r is **closed** then an analytic representation of such a disk contains the less than or equal to sign:

$$x^2 + y^2 \leq r^2$$

and if such a disk is **open** then an analytic representation of such a disk contains the strictly less than sign only:

$$x^2 + y^2 < r^2$$

The notion of **a boundary** is developed and defined in the study of Topology in a formal way which is fairly elaborate and rather pesky and we will not get into these details here.

At an intuitive level and in plain English, in the context of this book, it is enough to think about a circle as a boundary of a disk and conversely. A disk is the inner portion of a circle, give or take.

A disk is open if and only if it contains the bounding circle.

A disk is closed if and only if it does not contain the bounding circle. Thus, the disk that is shown in Figure 2.7 above is open.

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In the above definitions, for demonstration purposes, the angular displacement θ varies between zero inclusive and 2π exclusive:

$$\theta \in [0, 2\pi)$$

for both a circle and a disk.

The linear or the radial displacement ρ of a point that belongs to a disk varies between zero inclusive and r , the radius of that disk, **inclusive** for a closed disk:

$$\rho \in [0, r]$$

and **exclusive** for an open disk:

$$\rho \in [0, r)$$

In general, a circle is not a disk and a disk is not a circle unless both of these objects degenerate into a point, in which case they become indistinguishable.

A similar relationship and distinction exists between **a sphere** and **a ball** in three-space. A sphere is a boundary of a ball and conversely. A ball is the inner region of a sphere, give or take.

If a ball is open then it does not contain the bounding sphere and if a ball is closed then it does contain the bounding sphere.

An analytic representation of an origin-centered sphere with radius r in a Cartesian coordinate system is given by:

$$x^2 + y^2 + z^2 = r^2$$

where the equal sign is of significance.

The corresponding ball that is open is represented by:

$$x^2 + y^2 + z^2 < r^2$$

and if such a ball is closed then it is rendered symbolically as:

$$x^2 + y^2 + z^2 \leq r^2$$

and so on.

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But, as we promised, do not worry about all these details because we will not be relying on them in this book.

However, it pays to have firm mental grip on what a circle is, what a disk is and what is the difference between the two because the main character of this book is a circle.